Group Representation Theory Notes

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§1 Basics of Representation theory

Definition 1.1 (Group representation)**.** A *representation* of a group G is a morphism $\rho : G \to \mathrm{GL}(V)$, where V is a certain vector space. Commonly one refers to V as being the representation of G, taking as granted from the context the mosphism ρ.

Analogously a representation of a ring or a field is a morphism from it to an arbitrary $\mathrm{GL}(V)$. As it's usual with actions, if ρ is clear from the context we write

$$\rho_V(g)v = g \circ v = gv$$

with a slight abuse o notation.

Definition 1.2 (Group algebra)**.** Given a field k and a group G the *group algebra* of G is the set

$$kG := \left\{ \sum_{g \in G} \lambda_g g : \lambda_g \in k \right\}$$

equipped with the obvious sum (as a vector space) and the distributive multiplication that satisfies the multiplication of G.

We should prove that the one above is actually an algebra, but it is quite easy indeed. Given a representation ρ of G we can extend it linearly to a representation $\tilde{\rho}$ such that

$$\tilde{\rho} : kG \to \mathrm{End}(V)$$

and this is a ring representation (we go in End because multiplication in kG is not necessarily invertible).

Definition 1.3 (Algebra)**.** Given a field k, an *algebra* over it is a k-vector space equipped with a bilinear, unitary and associative product.

Just as with groups we define a representation of an algebra as a morphism $\rho : A \to \mathrm{End}(V)$.

Example 1.4 (Trivial representation)

For any group the map $\rho : G \to \mathrm{GL}(V)$ such that

$$g \mapsto 1$$

is the so called *trivial representation* of G.

Definition 1.5 (Regular representation)**.** For any group G, if we take $V = kG$ we get the so called *regular (left) representation* of G as

$$h \circ \sum_{g \in G} \lambda_g g \mapsto \sum_{g \in G} h\lambda_g g$$

this is equivalent to postulate that G acts on kG as the standard left multiplication $h \circ g := hg$ on elements of the form $g \in kG$, and then on all the other elements we extend it linearly.

Example 1.6 (Representation of a permutation)
For $G = S_n$ and $V = k^n$ the map that permutes the canonical basis is a representation, in other words

$$g \circ e_i := e_{g(i)}$$

is a representation.

Example 1.7 (Determinant)
Given a representation $\rho : G \to \mathrm{GL}(V)$, with V a vector space on k, the map

$$\det \circ \rho : G \to \mathrm{GL}(V) \to k$$

is also a representation (in particular is always 1-dimensional).

§1.1 Morphisms of representations

Definition 1.8 (Morphisms of representations)**.** Given a group G and two representations ρ_V and ρ_W over the same field k, a *morphism* between these is a linear map $T : V \to W$ such that is compatible with both ρ_V and ρ_W; that is, we have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle \rho_V(g)}\downarrow & & \downarrow{\scriptstyle \rho_W(g)} \\ V & \xrightarrow[T]{} & W \end{array}$$

for any $g \in G$. Equivalently we must have

$$\rho_W(g) \circ T = T \circ \rho_V(g).$$

These are also called *intertwined operators* or *G-equivariant maps*, or *G-morphisms*... they have a lot of names. Anyway we say that a morphism of representations is an *isomorphism* if it is invertible.

What we are saying essentially is, being that $\rho_V(g)$ and $\rho_W(g)$ are matrices, T commutes with all such matrices in the way given above. The set of all intertwined operators is oftend denoted as

$$\mathrm{Hom}(V, W) \geq \mathrm{Hom}_G(V, W)$$

which is always a non empty vector space (!).
Let's now consider two representations V and W, we can investigate what happens to the action of G when we take $V \oplus W$, as we expected

$$g \circ (v, w) = (g \circ v, g \circ w)$$

and no one is surprised. As usual with vector spaces, given two finite dimensional representations V, W, if we choose basis $\mathcal{B}_V$, $\mathcal{B}_W$ theese induce a morphism of representations from V and W to

$$\rho_V : G \to \mathrm{GL}_n(k)$$

and

$$\rho_W : G \to \mathrm{GL}_m(k)$$

in this way $\rho_{V\oplus W}(g)$ has also a nice matrix representation

$$\rho_{V\oplus W}(g) = \left(\begin{array}{c|c} \rho_V(g) & 0 \\ \hline 0 & \rho_W(g) \end{array}\right).$$

Lemma 1.9 (G-equivariant is equivalent to kG-equivariant)
Given a map between two representations V and W of G, we have that this is G-equivariant iff is kG equivariant, given k the field over which V is defined.

Proof. If T is G-equivariant we have that for any $g \in G$ one has

$$T\rho_V(g) = \rho_W(g)T$$

and so extending these to $\tilde{\rho}_V$ and $\tilde{\rho}_W$ we have that, given a certain

$$\sum_{g\in G} \lambda_g g \in kG$$

by linearity of T and $\tilde{\rho}$ we have

$$T\tilde{\rho}_V\left(\sum_{g\in G}\lambda_g g\right) = \tilde{\rho}_W T\left(\sum_{g\in G}\lambda_g g\right)$$

which means that T is kG-equivariant. On the other hand if T is kG-equivariant since $G \subseteq kG$ we have that T is also G-equivariant, and hence we have our thesis. □

Example 1.10 (Dual representation)
Given a representation V of G, it's dual space V^* has naturally a representation structure

$$\left(\rho_{V^*}(g)f\right)(v) = f\left(\rho_V(g)^{-1}v\right)$$

where the inverse is taken to make sure that it is an homomorphism.An equivalent formulation comes with the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{f} & k \\ {\scriptstyle \rho_V(g)}\downarrow & \nearrow_{\rho_{V^*}(g)\circ f} & \\ V & & \end{array}$$

a more compact way to write it is $g \circ f = f \circ g^{-1}$. An even more compact way to see this is through matrices, choosing a basis B and it's dual basis B^* of V^* we see that the actions as matrices satisfy

$$\rho_{V^*}(g) = \left[\rho_V(g)^{-1}\right]^{\intercal}.$$

Definition 1.11 (Tensor product of representations)**.** Given two representations V and W of a group their *tensor product* is the space

$$V \otimes W := \langle\{v \otimes w : v \in V, w \in W\}\rangle$$

equipped with the action

$$\rho_{V\otimes W}(g)(v \otimes w) := \rho_V(g)v \otimes \rho_W(g)w.$$

Remark 1.12 (Canonical identification of tensor products)**.** Given spaces W and V^* there is always a canonical bijection

$$F : W \otimes V^* \hookrightarrow \mathrm{Hom}(V, W)$$
$$\sum c_{wf} w \otimes f \mapsto \sum c_{wf} f(v) w$$

which is equivalent to say that on pure tensors our map is just

$$w \otimes f \mapsto F(w \otimes f)(v) := f(v)w$$

and for all the other elements of $V \otimes W$ we extend it linearly. This is just a basis free and abstract way of saying that, given elements $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$ we always have

$$(\mathbf{a} \otimes \mathbf{b})\,\mathbf{c} := (\mathbf{a}\mathbf{b}^\intercal)\,\mathbf{c} = \mathbf{a}\,(\mathbf{b}^\intercal \mathbf{c}) =: \mathbf{a}\langle \mathbf{b}, \mathbf{c}\rangle$$

This gives us immediately that if V and W representations of G, so is $V^* \otimes W$. Since the canonical injection defined above is equivariant (prove it) we hence have that **$\mathbf{Hom}(V, W)$ is a representation of G**. What is the action of this? We have that

$$(g \circ f)(v) = \rho_W(g)\left(f\left(\rho_V(g^{-1})v\right)\right) = \left(\rho_W(g) \circ f \circ \rho_V(g)^{-1}\right)(v)$$

and so if $V = W$ this is just a conjugation of $f \in \mathrm{End}(V)$!

Remark 1.13 (★ Internal Automorphisms)**.** In a way, if we see a representation of G just as a $H \leq \mathrm{GL}(V) \subseteq \mathrm{End}(V)$ monomorphic to G, it is natural to think of the action of G on $\mathrm{Hom}(V, V) = \mathrm{End}(V)$ for any $h \in H$ and $f \in \mathrm{End}(V)$ as the conjugation:

$$h \circ f = hfh^{-1}$$

in other ways this is just an internal automorphism generated by the elements of the subgroup H of $\mathrm{GL}(V)$.

§1.2 Subrepresentations and Maschke

Definition 1.14 (Subrepresentation)**.** Given a representation V of an object G, a *subrepresentation* of V is a subspace $U \leq V$ such that is G-stable, that is

$$G \circ U \subseteq U$$

which means that acting with G on U always gets us to elements still in U.

Example 1.15 (Fixed point of permutations)

Given $G = S_n$ with it's permutation representation on $V = k^n$, a subrepresentation of V is

$$U = \langle e_1 + e_2 + \cdots + e_n \rangle$$

and in this case is much more than G invariant: is pointwise fixed by the action of G. A non pointwise fixed example is the space W with Cartesian equation

$$\sum_{i=1}^{n} x_i = 0$$

which, if $\mathrm{car}K \not| n$ is not pointwise fixed for every point, $w \notin G \circ w$ for any w.

An operation we haven't yet touched is the quotient of spaces, we want to investigate what happens if we take two representation $V \geq U$ of G and consider V/U. We claim that this is a representation with the action

$$\rho_{V/U}(v+U) := \rho_V(g)v + U$$

and we should check that this is actually well defined and is a homomorphism. What does this mean in matrix terms? Given a basis u_i of U and a completion of it to a basis of V, the matrix of $\rho_V(g)$ in this basis has generally the form

$$\rho_V(g) = \left(\begin{array}{c|c} \rho_U(g) & * \\ \hline 0 & \rho_{V/U}(g) \end{array}\right)$$

one of the big questions of representation theory, as always when we deal with matrices, is wether, given a certain representation V of a group G, we can express the action of G on V as acting through block diagonal matrices. From the formula above we see that if we can find a couple of spaces $U \oplus U'$ that are both G stable, we have achieved our goal.

Example 1.16 (Decomposition of a representation of V)
Taking $G = S_n$ and $V = k^n$, since we have

$$\langle e_1 + e_2 + \cdots + e_n \rangle \oplus \left\langle \left\{ \sum_{i=1}^{n} x_i = 0 : x \in k^n \right\} \right\rangle$$

and these are G-stable we have that this is a decomposition of V as a representation.

Remark 1.17 (Lattice of subrepresentations)**.** If V is a representation of G and $U, W \leq V$ are subrepresentations of G then $U \cap W$ and $U + W$ are subrepresentations of G too.

Definition 1.18 (Fixed points)**.** For every representation V of G we can define the set of *fixed points* of V as

$$V^G := \{v \in V : gv = v \quad \forall g \in G\}$$

which can also be seen as the sum of all the subrepresentations isomorphic to the trivial one

$$V^G = \sum_{G \circ u = \{u\}} \langle u \rangle$$

this implies that V^G has naturally a subrepresentation structure.

Lemma 1.19 (Fixed points of the Hom representation)
For any pair V, W of representations of G we have that

$$\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G.$$

seeing $\mathrm{Hom}(V, W)$ as a representation of G through the tensor product.

Proof. From the definition of $\mathrm{Hom}(V, W)^G$ we have

$$\mathrm{Hom}_G(V, W) = \left\{ f \in \mathrm{Hom}(V, W) : \rho_V(g) \circ f \circ \rho_W(g)^{-1} = f \right\}$$

but since

$$\rho_V(g) \circ f \circ \rho_W(g)^{-1} = f \iff \rho_V(g) \circ f = f \circ \rho_W(g)$$

that's just the definition of G equivariant maps, and so we have the thesis. □

Definition 1.20 (Projector)**.** Given a finite group G and a field k we define the *projector* of G in kG as

$$\tilde{P} := \sum_{g \in G} g \in kG$$

this is always an element of $Z(kG)$. Moreover if $\text{char } k \nmid G$ we can define

$$P := \frac{1}{|G|}\tilde{P}$$

which has the nice property that $P^n = P$ for any $n \in \mathbb{N}$.

Lemma 1.21 (Projection to V^G)
Let V be a representation of a group algebra G, then for any $v \in V$ we have

$$\tilde{\rho}_V(P)v \in V^G$$

where $\tilde{\rho}_V$ is the regular representation on a group algebra kG.

Proof. It follows form the fact that $gP = P = Pg$ for any $g \in G$ that

$$g \circ (P \circ v) = gP \circ v = P \circ v$$

and so this satisfies the definition of V^G. □

This implies that $\tilde{\rho}_V(P) : V \to V^G$ is exactly a projection on V^G.

Lemma 1.22 (Kernel and Image of equivariant maps)
For any G-equivariant maps $T : V \to W$ its kernel and image are always G-stable.

Proof. For the kernel, given a $g \in G$ and $v \in \text{Ker}\,(T)$ we wish to prove that

$$g \circ v \in \text{Ker}\,(T)$$

on the other hand, since T is equivariant we have

$$T(g \circ v) = g \circ T(v) = g \circ 0 = 0$$

which is exactly what we wanted. On the other hand if $w \in \text{Im}\,(T)$ we have that exists a $v \in V$ for which we have

$$Tv = w$$

but hence for any $g \in G$ we get

$$g \circ Tv = T\,(g \circ v) \in \text{Im}\,(T)$$

and we have that also $\text{Im}\,(T)$ is G-stable. □

this lemma tells us once more that V^G is a subrepresentation of V.

Definition 1.23 (Irreducible representation)**.** A representation V of G is *irreducible* if its only subrepresentations are itself and the trivial one.

Definition 1.24 (Completely reducible)**.** A representation V of G is said to be *completely reducible*, or *semisimple* if it can be written as a direct sum of irreducible representations.

Example 1.25 (1-dimensional representations)

Every 1-dimensional representation is irreducible, clearly.

Example 1.26 (A non irreducible representation)

Take C_3 with the field $\mathbb{R}$, then its regular representation $\mathbb{R}C_3$ is not irreducible, since we have the subrepresentation

$$U = \left\{a1 + bg + cg^2 : a + b + c = 0\right\}.$$

We try and think to finite dimensional representations which are not completely reducible, but not many pop up to our minds. This makes us wonder if they even exists, indeed for finite groups we have the strong fact result that is Maschke's Theorem:

Theorem 1.27 (Maschke)

Let G be a finite group such that $\operatorname{char} k \nmid |G|$. Let V be a representation of G. Then any subrepresentation of G has a G-stable complement, so as a consequence every finite dimensional representation of G is completely reducible.

Proof. Let $U \leq V$ be G-stable. If W' is such that

$$V = U \oplus W'$$

let $\pi_U^{W'} =: \pi \in \operatorname{Hom}(V, U)$ be the projection onto U along W'. We know that G acts on $\operatorname{Hom}(V, U)$ let's take the projector P of G, then we have that

$$\operatorname{Hom}(V, U)^G = \operatorname{Hom}_G(V, U)$$

and so composing $P \circ \pi$ we get a G-equivariant map. By definition of P we have

$$P \circ \pi = \frac{1}{|G|} \sum_{g \in G} g \circ \pi = \frac{1}{|G|} \sum_{g \in G} \rho_U(g) \circ \pi \circ \rho_V(g)^{-1}$$

where at the RHS the $\circ$ represents function composition and on the LHS represents action. How does this function behave on U? If $u \in U$ we have

$$(P \circ \pi)u = \frac{1}{|G|} \sum_{g \in G} \rho_U(g) \circ \pi \circ \rho_V(g)^{-1}(u) = \frac{1}{|G|} \sum_{g \in G} g^{-1} \circ g \circ u = u$$

so $P \circ \pi_{|U} = \mathrm{id}_{|U}$ but then since $P \circ \pi$ is equivariant (is the composition of equivariant maps), it's kernel must be G-stable, since the kernel is just W', we are done. □

Corollary 1.28 (Strong Maschke)

For any representation V of G we have that the following are equivalent:

- Every subrepresentation of V has a stable complement.
- For any representation V we have irreducible subrepresentations V_i such that

$$V = \bigoplus_{i=1}^{k} V_i.$$

- For any representation V we have irreducible subrepresentations V_i such that

$$V := \sum_{i=1}^{k} V_i.$$

In particular Maschke's theorem tells us that the first condition is true, and so all the other ones must follow as well.

§1.2.1 Schur's Lemma

Another strong result about irreducible representations is Schur's lemma:

Lemma 1.29 (Schur)

For any G-equivariant map ϕ, which is not identically zero, between representations V and W of G we have the following:

(i) If V is irreducible then ϕ is injective.

(ii) If W is irreducible then ϕ is surjective.

Thus if both V and W are irreducible we have that ϕ must be an isomorphism, and hence the space of isomorphism between any two irreducible representations $\mathrm{Hom}_G(V, W)$ is either the null set (the set containing just the null element) or an algebra where every element can be inverted!

Proof. In the first case we have that

$$\mathrm{Ker}\,(\phi) \leq V$$

is a subrepresentation of V, but since ϕ is not identically zero we must have that $\mathrm{Ker}\,(\phi) = \{0\}$, and hence ϕ is injective. On the other hand in the second case we have

$$\mathrm{Im}\,(\phi) \leq W$$

is a subrepresentation of W, and since ϕ is not identically zero we must have $\mathrm{Im}\,(\phi) = W$ which means that ϕ is surjective. □

In the case of algebraically closed fields this result is much much stronger:

Corollary 1.30 (Schur's lemma for algebraically closed fields)

Le V, W be irreducible representations of G, if k is algebraically closed then $\mathrm{End}_G(V) = k\mathrm{id}_V$.

Proof. [1] For the first part we simply have that iff there exists a nonzero $f \in \mathrm{Hom}_G(V, W)$ then we have $\mathrm{Ker}\,(f) \leq V$ is a subrepresentation of V, which means that $\mathrm{Ker}\,(f) = 0$. Thus we have f injective, which means that

$$V \cong \mathrm{Im}\,(f) \leq W$$

which by irreducibility implies $V \cong W$. For the second part we have that, given a $f \in \mathrm{End}_G(V)$ for the algebraic closeness of k we always have an eigenvalue of f, let's call it λ. But this means that

$$0 \neq \mathrm{Ker}\,(f - \lambda I_V)$$

is G-stable, so it must be V. □

Corollary 1.31 (Triviality of action on irreducible representations)

If k is algebraically closed then any $z \in Z(kG)$ acts as a scalar on any irreducible representation.

Proof. contenuto... □

§1.3 Group functionals and Characters

Definition 1.32 (Class function)**.** For any group G and field k a *group functional* from G to k is just a function $f : G \to k$, the set of all such functions is denoted with $F(G, k)$. The set of group functionals which are constant over the conjugacy classes of G is denoted as $F_c(G, k)$.

With Schur's lemma we saw that working on an algebraically closed field and decomposing our space in

Definition 1.33 (Character)**.** Let V be a representation of G, then the *character* of V is

$$\begin{aligned} \chi_V : G &\to k \\ g &\mapsto \mathrm{tr}(\rho_V(g)) \end{aligned}$$

which is invariant up to G-equivariant maps. This can be naturally extended to kG, and is commonly denoted ad $\tilde{\chi}_V$.

Remark 1.34 (Character is equal on isomorphic representations)**.** For any equivariant isomorphism $T : V \to W$ we have

$$\rho_W(g) = T\rho_V(g)T^{-1}$$

and so $\chi_V(g) = \chi_W(g)$ for any $g \in G$. This means that on any pair of isomorphic representations, the character is the exact same function!

[1] This proof works only for finite dimension, although the result works for general representations as well.

Proposition 1.35 (Properties of the character)
For any V, W representations of G we have:

1. The character is a class function
2. $\chi_{V\oplus W} = \chi_V + \chi_W$
3. $\chi_{V\otimes W} = \chi_V \chi_W$.

Proof. The first formula is easy to see from the matrix interpretation of □

Let's now specialize to the case where $k = \mathbb{C}$, if we have that $|G| < \infty$ this means that for $M := \exp G$ we have

$$g^M = 1$$

and so it's associated action is $\rho_V(g)$ on an irreducible representation is a scalar, in particular is an M-th root of unity. Since for any matrix $\rho(g)$ we have that

$$\operatorname{tr}(\rho_V(g)) = \chi_V(g) = \dim V \iff \rho_V(g) = I$$

$$\operatorname{Ker}(\rho_V) = \{g \in G : \chi_V(g) = \dim V\}$$

Definition 1.36 (Inner product on group functionals)**.** On $F(G, \mathbb{C})$ and $F_C(G, \mathbb{C})$ we define an inner product

$$(f, f') := \frac{1}{|G|} \sum_{g\in G} f(g)\overline{f'(g)} = \overline{(f', f)}.$$

Proposition 1.37 (First Orthogonality relation)
If $k = \mathbb{C}$ and $|G| < \infty$, given two representations V, W we have that

$$(\chi_V, \chi_W)_G = \dim \operatorname{Hom}_G(V, W)$$

and because that number is purely real, in particular is an integer, it's also equal to $(\chi_W, \chi_V)_G$.

Proof. We want to compute

$$(\chi_V, \chi_W)_G = \frac{1}{|G|} \sum_{g\in G} \chi_V(g)\overline{\chi_W(g)}$$

since we are on $\mathbb{C}$ the character of the dual representation is just the conjugate, and so we have

$$\cdots = \frac{1}{|G|} \sum_{g\in G} \chi_{V\otimes V^*}(g)$$

so through linearity we get

$$\doteq \chi_{V\otimes W^*}\left(\frac{1}{|G|}\sum_{g\in G} g\right) = \chi_{V\otimes W^*}(P) = \operatorname{tr}\left(P_{|V\otimes W^*}\right)$$

but the trace of a projection is just the dimension of the image, so since P is a projection from $\operatorname{Hom}(V, W)$ to $\operatorname{Hom}_G(V, W)$ we have our result. □

So putting together the result of Maschke and orthogonality of characthers we get that the set $\mathrm{Irr}(G)$ of irreducible representations of G are always

$$|\mathrm{Irr}(G)| \leq \dim F_C(G, \mathbb{C})$$

and so we have that any χ_V is a $\mathbb{Z}_{\geq 0}$-linear combination of χ_{V_i}.

Corollary 1.38 (Fourier Coefficients of the characters)

If $\{V_i, i \in I\}$ is a set of isomorphsim classes of irreducible representations of G we have that any V representation of G is

$$V = \bigoplus_{i=1}^{k} V_i^{\oplus n_i}$$

moreover we have

$$(\chi_V, \chi_i)_G = n_i$$

so the n_i are the fourier coefficients of χ_V in the orthonormal basis χ_i. This means in particular that the multiplicities n_i are unique.

Proposition 1.39 (Equality of Characters implies equality of representation)

Let (V, ρ) and (W, ρ') be two representations of a group G over complex numbers. If $\chi_\rho = \chi_{\rho'}$ then $V = W$.

Proof. Let's consider the set of all irreducible representations of G as

$$\mathrm{Irr}_G(G) = \{V_1, \dots V_r\}$$

now by Maschke's theorem we can decompose both W and V as a sum of irreducibles:

$$V = \bigoplus_{j=1}^{r} V_i^{\oplus n_i}$$
$$W = \bigoplus_{j=1}^{r} V_i^{\oplus m_i}$$

since the character of a sum of representations is the sum of the character, denoted with χ_i the characters of V_i we have that

$$\chi_\rho = \sum_{i=1}^{r} n_i \chi_i$$

but since the character are the same we have that

$$\sum_{i=1}^{r} n_i \chi_i = \sum_{i=1}^{r} m_i \chi_i$$

and so taking the inner product on both sides wrt to any χ_i we have

$$n_i = (\chi_\rho, \chi_i) = (\chi_{\rho'}, \chi_i) = m_i$$

and hence $V = W$ as desired. □

Corollary 1.40 (Characterisation of irreducibility)

If V is a finite dimensional representation of G on $\mathbb{C}$ then V is irreducible if and only if

$$(\chi_V, \chi_V) = 1.$$

Proof. We decompose V as a sum of irreducible representations again

$$V = \bigoplus_{i=1}^{r} V_i^{\oplus n_i}$$

from this we have a very explicit formula for χ_V, which we can use to compute the quantity we are considering

$$(\chi_V, \chi_V) = \left(\sum_{i=1}^{r} \chi_i n_i, \sum_{i=1}^{r} \chi_i n_i \right).$$

If that expression is 1 we see immediately that we must have $n_j = \delta_{ij}$ for some $i \in \mathbb{N}$, and hence V is irreducible. The other side of the thesis follows immediately from first orthogonality relations, and so we are done. □

Corollary 1.41 (V is irreducible iff V^* is irreducible)

Let V be a finite dimensional complex representation of a finite group G then V is irreducible if and only id V^* is irreducible.

Proof. We have the following chain of equalities

$$1 = (\chi_V, \chi_V) = \overline{(\chi_V, \chi_V)} = (\overline{\chi_V}, \overline{\chi_V}) = (\chi_{V^*}, \chi_{V^*})$$

hence we have proven that V^* is irreducible by the previous result. The converse follows from identifying V^{**} with V and invoking the finite dimensionality of V. □

Lemma 1.42 (Action of the regular representation)

For any finite G acting on $\mathbb{C}G$ through the regular representation we have that

$$\chi_{\text{reg}}(g) = |G|\, \delta_{eg}$$

where e is the identity.

Proof. It's clear that the character if $g = e$ the matrix we are considering is the identity, and so we have our thesis.
The difficult case is when $g \neq e$. In the action of g on $\mathbb{C}G$ we see that

$$e_h \mapsto e_{gh}$$

and so in the basis $\{e_g : g \in G\}$ every column of the matrix of $\rho(g)$ consists of just a non zero entry, equal to 1. Moreover we can't have that if $h_1 \neq h_2$

$$e_{gh_1} = e_{gh_2}$$

since acting left and right by g^{-1} we have $h_1 = h_2$. This mean that $\rho(g)$ is just a permutation matrix, and since $g \neq e$ we have

$$e_{gh} \neq e_h$$

for any h, this means that along the main diagonal there are only zeros, and hence the theorem follows. □

Theorem 1.43 (Decomposition of the regular representation)

Let $V_1, \dots, V_k$ be the complete set of irreducible representations of G and let χ_i be the corresponding characters. Then the regular representation decomposes as

$$\mathbb{C}G = \bigoplus_{i=1}^{k} V_i^{\oplus \dim V_i}.$$

Proof. Let's calculate the fourier coefficients of χ_{reg} using the previous lemma:

$$(\chi_i, \chi_{reg}) = \frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_{reg}(g)} = \frac{1}{|G|} |G| \, \delta_{eg} \chi_i(e) = \dim V_i$$

which by uniqueness of the decomposition in irreducible is our thesis. Explicitly we have that if

$$V = \bigoplus V_i^{\oplus n_i}$$

then computing the inner product using this formula and the additivity of characters we gain

$$\dim V_i = (\chi_i, \chi_{reg}) = \left(\chi_i, \sum n_i \chi_i\right) = n_i$$

and hence we are done. □

Corollary 1.44 (Sum of squares formula)

For any group G, given the set of irreducible representations V_i we always have that

$$|G| = \sum_{i=1}^{k} (\dim V_i)^2 .$$

Proof. Taking the dimension on both sides of the formula

$$\mathbb{C}G = \bigoplus_{i=1}^{k} V_i^{\oplus \dim V_i}$$

we have the thesis. □

§1.4 Commutators and abelian groups

Let's consider a finite group G of order d. Given a representation

$$\rho : G \to \mathrm{GL}(V)$$

we have that for every $g \in G$ this is a matrix. Moreover by Lagrange's theorem we have that

$$g^d = 1 \implies \rho(g)^d = \mathrm{id}$$

this means that denoted with $X := \rho(g)$ its minimal polynomial m_X satisfies

$$m_X \mid x^d - 1$$

now we make two important assumption about our field:

1. All the roots of $x^d - 1$ exist in k.
2. We have $(\text{ char } k, d) = 1$ (this is to make sure that we have no repeated roots in $x^d - 1$).

Then we have that m_X factors as a product of distinct linear factors, and we have diagonalizability! All the above assumptions are naturally satisfied if we work on $\mathbb{C}$, which is anyway the field where we'll be working on. We need here an ounce of linear algebra

Proposition 1.45 (Simultaneous diagonalizability)
Let $X, Y \in \mathrm{End}(V)$ diagonalizable. Then X and Y are simultaneously diagonalizable iff they commute.

Proof. If they are simultaneously diagonalizable the thesis clearly holds, since all diagonal matrices commute. To prove the converse we decompose V in X eigenspaces

$$V = \bigoplus E_\lambda(X)$$

now if we apply Y to any $E_\lambda(X)$ we stay in $E_\lambda(X)$, in fact if $v \in E_\lambda(X)$ we have through commutativity

$$X(Yv) = YXv = Y\lambda v = \lambda Yv$$

which just implies $Yv \in E_\lambda(X)$. But this immediately concludes, because

$$X_{|E_\lambda(X)} = \lambda \mathrm{id}$$

and so since Y is again diagonalizable in $E_\lambda(X)$ if we just pick any basis of $E_\lambda(X)$ that diagonalizes Y we get for free that X is diagonal in that same base (since it is diagonal in any base!). □

Putting together theese two results we have that if A is a finite abelian group, then for any representation V we can pick a basis of V where all $\rho(a)$ are diagonal matrices! A fun little application is:

Corollary 1.46 (Representation of finite ciclyc groups)
For any $r \in \mathbb{N}$ and $d \in \mathbb{N}$ we have that if

$$\left(\mathbb{Z}/d\mathbb{Z}\right)^r \hookrightarrow \mathrm{GL}_n(\mathbb{C})$$

is an injective morphism then $r \leq n$.

Let A be an abelian finite group and $\rho : A \to \mathrm{GL}(V)$. If each $\rho(g) \in \mathrm{GL}(V)$ is diagonalizzable then all elements $\rho(g)$ are diagonalizable, so if we take

$$V_i := \langle v_i \rangle$$

where v_i is the basis that diagonalizes all the $\rho(g)$. For any $g \in A$ and $i \leq \dim V$ then we have that

$$\rho(g)V_i \subseteq V_i$$

which means that for all i the V_is are subrepresentations of V. But it's clear that all $\rho(g)$ are diagonalizable, since we saw earlier that they all must satisfy

$$[\rho(g)]^{|G|} - 1 = 0$$

which we have assumed has all distinct roots in k. From this discussion we evince

Proposition 1.47 (Irreducible representations of finite abelian groups)
All the irreducible representations of a finite abelian group A are the finite dimensional ones. Moreover from the sum of squares formula we have that there are exactly $|A|$ irreducible representations of A.

From the structure theorem of finite abelian groups we know that any finite abelian A decomposes as

$$A \cong C_{p_1^{e1}} \times C_{p_2^{e2}} \times \cdots \times C_{p_k^{ek}}$$

where p_i are not necessarily distinct primes. This means we can find $g_1, \ldots, g_k$ elements of A (ideally the elements $(a_1, 1, \ldots, 1)$, $(1, a_2, \ldots, 1)$, $\ldots$, $(1, 1, \ldots, a_k)$, generators of the canonical isomorphic images of $C_{p_i^{e_i}}$ in A) of order respectively $p_i^{e_i}$. Then choosing the representation of A amounts to just choosing $\rho(g_i)$ for all g_i. For any of these elements we have $p_i^{e_i}$ choices, and since this choices are all independent the number of distinct representations of A are $p_1^{e_1} p_2^{e_2} \ldots p_k^{e_k}$, which is exactly what we concluded in the previous result.

Definition 1.48 (Commutator)**.** For any group G the function $G \times G \to G$ such that

$$[g, h] := ghg^{-1}h^{-1}$$

is called the *commutator* of g and h. The *derived subgroup* of G (also called the commutator group) is defined to be

$$[G, G] := \langle [g, h] : g, h \in G \rangle$$

which is a normal subgroup of G, the *abelianization* of G is the group $G/[G, G]$.

We immediately see that for any one dimensional representation $\rho : G \to V$ of any finite group G we have that

$$\rho([g, h]) = 1$$

since the image is commutative. From a generalization of the first homomorphism theorem we evince immediately that

$$\begin{array}{ccc} G & \xrightarrow{\rho} & \mathbb{C}^* \\ {\scriptstyle\pi}\downarrow & \nearrow_{\overline{\rho}} & \\ G/[G,G] & & \end{array}$$

and so immediately we can decompose any representation of any group G in a product of two morphisms, one that is just a representation of an abelian group, and the other one which is a group morphism.

Corollary 1.49 (Number of monodimensional representations)
If k is algebraically closed and (char $k, |G|) = 1$ we have that the number of finite dimensional representations of G is $|G : [G : G]|$.

Proof. Follows from the number of 1-dimensional representations of abelian groups. □

Example 1.50 (Symmetric one dimensional representations)
How many one dimensional representations of S_n are there?

Solution. This boils down to determining the representation of the abelianization of S_n. Since for any $g_1, g_2 \in S_n$ we have

$$\text{sign}([g_1, g_2]) = \text{sign}(g_1)\text{sign}(g_2)\text{sign}(g_1)^{-1}\text{sign}(g_2)^{-1} = 1$$

this means that

$$[G, G] \leq A_n$$

□

§1.5 Character tables

Let's return to class functions. For any $f \in F_c(G, \mathbb{C})$ and any $\rho : G \to \text{GL}(V)$ we can define an action of f on V in the following way

$$s_f := \sum_{g \in G} f(g)\rho(g) \in \text{Hom}_{\mathbb{C}}(V, V).$$

This has some quite nice properties, for example we have that

$$s_1 = \sum_{g \in G} \rho(g)$$

which is basically the projector multiplied by $|G|$.

Remark 1.51. The projector P has the property that

$$\rho(h)(Pv) = Pv$$

for any $v \in V$, and it really is the projection transformation onto the space of fixed points

of V, that is V^G.

Actually the property of commutation enjoyed by the projector is much more general:

Lemma 1.52 (Commutativity of class functions)
For any $f \in F_c(G, \mathbb{C})$ and $h \in G$ we have that

$$\rho(h)s_f = s_f\rho(h)$$

that is, s_f is G-equivariant.

Proof. An explicit calculation, we have

$$\rho(h)\left(\sum_{g\in G} f(g)\rho(g)\right)\rho(h)^{-1} = \sum_{g\in G} f(g)\rho(hgh^{-1}) = \sum_{g\in G} f(g)\rho(g)$$

which proves our thesis. □

Proposition 1.53 (Trace and inner product formula)
For any $f \in F_c(G, \mathbb{C})$ we have that

$$\chi_\rho(s_f) = \sum_{g\in G} f(g)\chi_\rho(g).$$

Proof. Follows from the linearity of trace. □

From this last property follows that for any $f \in F(G, k)$ one has

$$|G|\,(f, \overline{\chi}_\rho) = \chi_\rho(s_f).$$

Theorem 1.54 (Orthonormality of characters)
Let $\chi_1, \chi_2, \dots, \chi_n$ be all the irreducible characters of a group G (that is, the characters associated with any irreducible representation of G), then this is an orthonormal base of $F_c(G, \mathbb{C})$.

Proof. We already proved that the characters are an orthonormal set, in fact

$$(\chi_i, \chi_j) = \delta_{ij}$$

it remains to be shown that these span all $F_c(G, k)$. Let's consider $f \in F_c(G, \mathbb{C})$ such that

$$f \in H^\perp$$

where

$$H = \langle \chi_1, \dots, \chi_k \rangle$$

this is equivalent to saying that for all χ_i we have

$$(f, \chi_i) = 0 \implies (f, \overline{\chi}_i) = 0$$

but we proved before that

$$(f, \overline{\chi}_i) = \frac{\chi_i(s_f)}{|G|}$$

now by schur's lemma we now have that (since we are working on a closed field), since s_f is an equivariant map, over any V_i irreducible representation associated with χ_i we have that s_f acts as a scalar on every irreducible representation of G, but since this scalar k is just given by

$$\chi_i(s_f) = k \dim_{\mathbb{C}} V_i$$

we get that s_f is identically on any V_i, which implies that $f = 0$, as desired. □

Corollary 1.55 (Conjugacy classes of a group)

The number of conjugacy classes of G equal the number of it's irreducible representations.

Proof. Since the dimension of $F_c(G, \mathbb{C})$ is just the number of conjugacy classes of G, we have that k, the number of irreducible characters, is equal to the dimension of $F_c(G, k)$. □

Example 1.56 (Dihedral group)

For a dihedral group D_4 we have that the group of commutators is

$$[D_4, D_4] = \langle r^2 \rangle$$

where r is an element of order 4. This means that the number of one dimensional representations of D_4 is just 4. From the sum of squares formula we have

$$8 = 1^2 + 1^2 + 1^2 + 1^2 + \dots$$

what we are missing is a four, but since we have just listed all one dimensional representation of G that four must be written as a 2^2, which means that all the irreducible representations of D_4 are just the 4 one dimensional one and the 2 dimensional one. This implies that D_4 has exactly 5 characters (or 5 conjugacy classes).

We can express the characters of a group through a **character table**, for example for D_4 we have:

	$\{e\}$	$\{r^2\}$	$\{r, r^3\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	-1	-1	1
χ_4	1	1	1	-1	-1
χ_5	2	-2	0	0	0

in every cell a_{ij} we have the value that the character χ_i assumes on the class j. The first character is just the one associated to the trivial representation.
For the group of quaternions Q_8 on the other hand we have that

	$\{1\}$	$\{-1\}$	$\{i, i^3\}$	$\{j, j^3\}$	$\{k, k^3\}$
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	-1	-1	1
χ_4	1	1	1	-1	-1
χ_5	2	-2	0	0	0

and so we have two different nonisomorphic groups that have the same character table.

§1.6 Algebraic integers and Frobenius's divisibility theorem

Definition 1.57 (Algebraic integer)**.** A comples number α is called an *algebraic integer* if it is a root of some monic polynomial with integer coefficients.

This definition is basically equal to the one of an algebraic number, except for the restriction on the first coefficient of the polynomial.

Proposition 1.58 (Algebraic integers aren't rational)
If we denote with A the set of algebraic integers then we have

$$\mathbb{Q} \cap A = \mathbb{Z}$$

Proof. We clearly have that

$$\mathbb{Z} \subseteq A \cap \mathbb{Q}$$

then we just need to show the other inclusion. Let $\alpha \in \mathbb{Q} \cap A$, we can write

$$\alpha = \frac{p}{q}$$

with p, q coprime positive integers. Since α is algebraic there must exist finite $a_1, \dots, a_n$ integers such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

multiplying both sides by q this becomes a complete polynomial in p, q:

$$p^n = -a_{n-1}p^{n-1}q - \cdots - a_1pq^{n-1} - q^n$$

but this implies that

$$q \mid p^n \implies q \mid p$$

which means that $q = 1$, as desired. □

Proposition 1.59 ($\mathbb{Z}$-modules are stable by algebraic integers action)
The number α is an algebraic integer if and only if there is a nonzero abelian finitely generated subgroup $M \subseteq \mathbb{C}$ such that $\alpha M \subseteq M$ (that is M is a module over the ring of algebraic integers).

Proof. We firstly prove that if α is an algebraic integer then we have the module condition. Since α is algebraic we have

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then let's consider the finitely generated abelian group

$$M := \langle \alpha^0, \alpha^1, \dots, \alpha^{n-1} \rangle_{\mathbb{Z}}$$

with complex coefficients, equipped with the addition structure. Then since

$$\alpha^n = -a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha - a_0$$

it's clear that multipliying any sum of powers of order $< n$ of α by α is again equal to some sum of powers of α of order $< n$, which means that $\alpha M \subseteq M$, as we wished to show.
Assume now that there exists a finitely generated $\mathbb{Z}$-module M that is α-stable. Let's prove that α is an algebraic integer: let $e_1, \dots, e_m$ be a finite set of generators of M then we must have, using Einstein convention, that there exist $b_{ij} \in \mathbb{Z}$ such that

$$\alpha e_i = b_{ij} e_j$$

but this means that α just acts as a linear operator on M. Since the composition of the action of α is just multiplication, then calling B the matrix of the coefficients b_{ij} we obtain by Hamilton-Cayley that

$$\det(\alpha I - B) = 0$$

which is a monic polynomial with integer coefficients having α as a root. □

Proposition 1.60 (Algebraic integers are a ring)
We have that for any $\alpha, \beta \in A$ the set of algebraic integers

- The opposite is still in the set $-\alpha \in A$.
- The sum is still in the set $\alpha + \beta \in A$.
- The product is still in the set $\alpha\beta \in A$.

Proof. The first condition is very easy to prove. We start then with the product: if α, β are algebraic then there exist two $\mathbb{Z}$-modules such that

$$\alpha M \subseteq M$$
$$\beta N \subseteq N$$

denoting with e_i and f_j the basis of these, then the product $MN := \langle e_i f_j \rangle$ is also a module, and is $\alpha\beta$ stable because

$$\alpha\beta x_{ij} e_i f_j = x_{ij} \alpha e_i \beta f_j \in MN$$

(where $x_{ij} e_i f_j$ in Einstein notation is a generic element of MN) and so $\alpha\beta$ is again algebraic. With the sum is pretty much the same, since MN is also $\alpha + \beta$ stable. □

The algebraic integers are for us a tool to show the following theorem

Theorem 1.61 (Frobenius divisibility theorem)

Let's consider a finite group G and V an irreducible representation of G, then we have

$$\dim V \mid |G|\,.$$

Proof. Let's call $\rho : G \to \mathrm{GL}(V)$ the morphism associated to the representation, and χ the character of (V, ρ). We have already seen that all the elements $\rho(g)$ are diagonalizable and that on the diagonal of each element appear only roots of unity. Since the roots of unity are by definition algebraic integers, this means that $\chi(g) \in A$ is an algebraic integer for any $g \in G$. Let $C_1, \dots, C_k$ be the conjugacy classes of G, denoting with r_i the cardinalities of each of these we define the elements of $\mathbb{C}[G]$ as

$$K_i = \sum_{g \in C_i} g.$$

Because the conjugacy classes are by definition stable by conjugation then

$$gK_i = K_i g$$

for any $g \in \mathbb{C}[G]$, hence $K_i \in Z(\mathbb{C}[G])$ for all i. Since this are a maximal set of linear independent elements of $Z(\mathbb{C}[G])$ they must be a basis. We've already proven that $K_i \in Z(\mathbb{C}[G])$ means that

$$\rho(K_i) = \alpha_i \mathrm{id}$$

where $\alpha_i \in \mathbb{C}$ is just some complex constant. This implies $\chi(K_i) = \alpha_i \dim V$ but using the linearity of the trace and invariance on conjugacy classes we get

$$r_i \chi(g_i) = \sum_{g \in C_i} \chi(g_i) = \chi(K_i) = \alpha_i \dim V.$$

We wish to show that the α_i are eigenvalues of some operator, from which will follow that they're also algebraic. Since

$$K_i K_j = c_{ij}^m K_m$$

with $c_{ij}^m \in \mathbb{Z}$ (it just follows from expanding the product). This means that on the $\mathbb{Z}$-module with α_i as generators then $\rho(K_i)$ is an action such that

$$\rho(\alpha_i)\alpha_j = c_{ij}^m \alpha_m$$

and hence we can write this, fixing i, in matrix form

$$M_i \vec{\alpha} = \lambda_i \vec{\lambda}$$

where $(M_i)_{jm} := c_{ij}^m$, and hence again by Cayley-Hamilton

$$\det(\alpha_i - M_i I) = 0$$

which shows that all the α_i are algebraic integers.
We are ready to conclude the proof, from the irreducibility of V we conclude

$$1 = (\chi_V, \chi_V) = \frac{1}{|G|} \sum_{g \in G} \chi(g)\overline{\chi(g)} \implies \sum_{g \in G} \chi(g)\overline{\chi(g)} = |G|$$

regrouping the terms by conjugacy class we obtain

$$\sum_{i=1}^{k} r_i \chi(g_i)\overline{\chi(g_i)} = |G|$$

substituting the terms we had found previuosly we find

$$\dim V \sum_{i=1}^{k} \alpha_i \chi(g_i) = |G| \implies \frac{|G|}{\dim V} = \sum_{i=1}^{k} \alpha_i \chi(g_i).$$

But we are done, because the RHS is a sum and product of algebraic numbers, and hence is an algebraic number, whereas the LHS is rational, and hence they both must be integer, concluding our proof. □

§2 Modules and representations

Definition 2.1 (Module)**.** Given a ring R with identity 1 then a *left R-module* consist of an abelian group $(M, +)$ and an operation $\cdot : R \times M \to M$ (a multiplication by a scalar basically) such that for any $r, s \in R$ and $x, y \in M$ we have

- $r(x + y) = rx + ry$.
- $(r + s)x = rx + sx$.
- $(rs)x = r(sx)$.
- $1 \cdot x = x$.

Basically we can think of a module as a vector space over a ring.

Clearly any G abelian group has naturally a $\mathbb{Z}$-module structure. Modules are connected to representations in the sense that given any $\rho : G \to \mathrm{GL}(V)$ we are equipping the space V with a module structure, in particular we are saying that (V, ρ) is a $k[G]$-module (or shortly a G-module), where the multiplication by a scalar for any $v \in V$ is

$$\left(\sum_{g\in G} a_g g\right) \cdot v := \sum_{g\in G} a_g \rho_V(g)v.$$

§2.1 Representations of the symmetric group

We'll denote as $\underline{n}$ the set $(n+1)^{(n+1)}$ of the permutations of n numbers.
For any element $\sigma \in S_n$ we define a vector

$$\sigma \mapsto (i_1, i_2, \dots, i_n)$$

where the i_j are the length of the j-th cycle (sorted by length) in the distinct cycle decomposition of σ. As an example if

$$\sigma = (1, 3, 5, 2, 6)(8, 9)(4)(7)$$

then it's associated vector is $(5, 2, 1, 1)$. By definition the elements of any such vector are always sorted in descending order.

Definition 2.2 (Partition)**.** For any $n \in \mathbb{N}$ a *partition* of n is a non increasing vector of natural positive numbers whose components sum up to n.

We want to study the representations of S_n because in any $\mathrm{GL}_n(k)$ the group of diagonal matrices D satisfies

$$N_{\mathrm{GL}_n(k)}(D)/D \cong S_n$$

which is quite crazy.

Definition 2.3 (Young diagram)**.** A *Young diagram* is a finite collection of boxes, arranged in left-justified rows, with the row lengths in weakly decreasing order.

An example of this is showed in figure 1. There exists a natural bijection between young diagrams consisting of n boxes and partitions of n. An example of this is Figure 1.

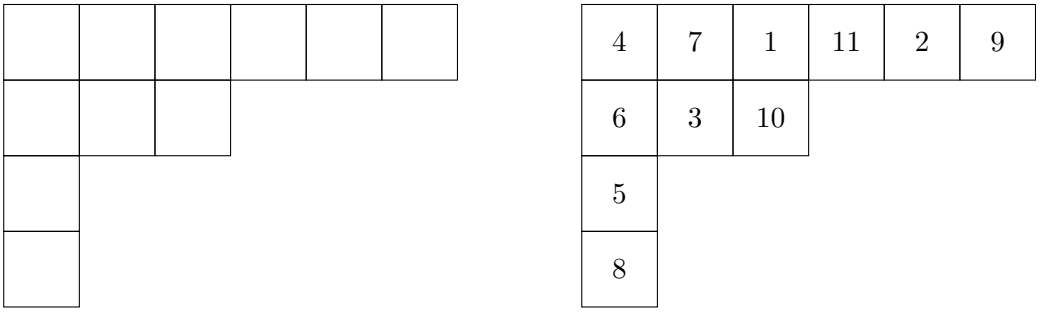


Figure 1: The unique Young diagram associated to the partition $(6, 3, 1, 1)$ of 11, and an associated Young Tableaux.

Definition 2.4 (Young Tableaux)**.** A *Young tableaux* is a filling of the boxes of a young diagram of shape λ (a partition of n) with the numbers from 1 to n. The set of all Young Tableaxues is denoted as YT. Given a partition λ of n we denote the set of all Young tableauxes associated to λ by YT_λ.

For any Young tableaux associated to a partition λ we can define a subset S_λ of n: the set of elements that make each row invariant of a Young tableaux of shape λ wrt the action of S_n.

Definition 2.5 (Row equivalence)**.** Given two Young tableauxes we say that these are *row equivalent* if on each row of the first young tableaux appear the same number as the ones appearing in the same of row of the second one.

Definition 2.6 (Young tabloid)**.** A row equivalence class of Young tableauxes is called a *Young tabloid*. The set of all Young tabloids of shape λ is denoted as YTD_λ.

The core idea is to let S_n act on YT_λ, by permuting the numbers in the tableaux applying them $\sigma \in S_n$. This gives a transitive action, which is not that interesting, but by taking the formal linear combinations $\mathbb{C}[YT_\lambda]$:

Lemma 2.7 (Young Tableauxes algebra structure)
For any partition λ we have that

$$\mathbb{C}[TY_\lambda] \cong \mathbb{C}[S_n].$$

Proof. We define an isomorphism choosing an arbitrary $t \in TY_\lambda$:

$$\begin{aligned} \varphi : \mathbb{C}[YT_\lambda] &\to \mathbb{C}[S_n] \\ \sigma(t) &\mapsto \sigma \end{aligned}$$

this is well defined for any element of TY_λ because of the transitivity of the action and is also easy to show that is equivariant. □